

Lecture 9

Why the integrals commute: the Toda ring is integrable

Objectives. Course map: the last step of G1, and the classical half of G5. [Lecture 8](#) found the conserved quantities of the Toda ring — the spectrum of its Lax matrix — and left a debt: conserved is not the same as in involution, and [Definition 5.5](#) asks for both. This lecture pays the debt. It packages all N integrals into one function of a spectral parameter, the transfer matrix, proves that its values Poisson-commute from a single identity satisfied by a single site, checks independence, and applies [Theorem 5.7](#): the periodic Toda chain is Liouville integrable, and its motion winds round tori that drift. The identity is the one [Lecture 14](#) quantises.

9.1 The question

The Toda ring of [Lecture 8](#) has N particles on a circle,

$$H = \frac{1}{2} \sum_{n=1}^N p_n^2 + \sum_{n=1}^N e^{q_n - q_{n+1}}, \quad q_{N+1} = q_1, \quad (9.1)$$

and its Lax matrix L has a spectrum that does not change in time ([Theorem 8.2](#)), so the N numbers $\text{tr } L^k$ are conserved. [Lecture 8](#) also rewrote the eigenvalue problem of L as a transport from site to site, $\Phi_{n+1} = L_n(\lambda)\Phi_n$, with the 2×2 site matrix

$$L_n(\lambda) = \begin{pmatrix} \lambda - p_n & e^{-q_n} \\ -e^{q_n} & 0 \end{pmatrix}, \quad \det L_n(\lambda) = 1, \quad (9.2)$$

which contains the variables of site n and nothing else.

[Definition 5.5](#) asks for more than conservation: N integrals that Poisson-commute with each other and are independent. Nothing in the Lax equation says anything about brackets between the $\text{tr } L^k$. So the question of this lecture is

are the Toda ring's integrals in involution and independent — and if so, what does its motion look like?

The plan has three steps. Build all the integrals at once, as the coefficients of one polynomial $t(\lambda)$ (§ 9.2). Show that $\{t(\lambda), t(\mu)\} = 0$ follows from one identity satisfied by one site matrix (§§ 9.3 to 9.5). Then independence and [Theorem 5.7](#) do the rest (§§ 9.6 and 9.7).

9.2 All the integrals at once: the transfer matrix

Transport all the way round the ring.

Definition 9.1 (Monodromy and transfer matrices). For a chain of N sites with site matrices $L_n(\lambda)$, the *monodromy matrix* and the *transfer matrix* are

$$T(\lambda) = L_N(\lambda) \cdots L_2(\lambda) L_1(\lambda), \quad t(\lambda) = \text{tr} T(\lambda). \quad (9.3)$$

The chain is *ultralocal* if variables at different sites Poisson-commute: $\{(L_n)_{ab}, (L_m)_{cd}\} = 0$ for $n \neq m$. When tensor legs are also in play (§ 9.3), the site moves to a superscript: $L^{(n)} = L_n$.

The Toda ring is ultralocal, since L_n involves only (q_n, p_n) . By (9.2), $\det T = 1$, and $T(\lambda)$ is what a solution of the recursion picks up in one circuit (Figure 9.1).

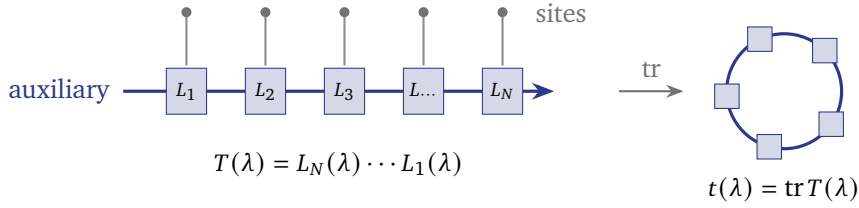


Figure 9.1: Left: the monodromy matrix as transport. One auxiliary line threads every site in turn, picking up each site matrix; the vertical stubs are the sites, which carry the dynamical variables. Right: the trace closes the auxiliary line into a loop with no ends to distinguish. That picture is the whole proof of Corollary 9.8.

What the transfer matrix contains

Expand the product (9.3) as a sum over paths through the index values 1, 2 of the 2×2 factors. The entry $(L_n)_{22} = 0$ forbids a path to stay at 2, so each visit to 2 enters through $(L_n)_{21} = -e^{q_n}$ and leaves at the next site through $(L_{n+1})_{12} = e^{-q_{n+1}}$: a factor $-e^{q_n - q_{n+1}}$ in place of two factors $\lambda - p$. The trace closes the path, so the pair may straddle the join. Hence

$$t(\lambda) = \sum_S \prod_{n \in S} (-e^{q_n - q_{n+1}}) \prod_{m \notin \bar{S}} (\lambda - p_m), \quad (9.4)$$

the sum over sets S of bonds, no two sharing a site, with $\bar{S}$ the sites they cover. Writing $t(\lambda) = \sum_{k=0}^N c_k \lambda^{N-k}$,

$$c_0 = 1, \quad c_1 = - \sum_n p_n, \quad c_2 = \sum_{i < j} p_i p_j - \sum_n e^{q_n - q_{n+1}}, \quad H = \frac{1}{2} c_1^2 - c_2. \quad (9.5)$$

So $t(\lambda)$ is a polynomial of degree N whose N non-trivial coefficients $c_1, \dots, c_N$ are functions on phase space, and H is one of their functions.

The same integrals as Lecture 8's

A number λ is an eigenvalue of Lecture 8's Lax matrix L exactly when its eigenvalue recursion has a solution with $\psi_{n+N} = \psi_n$. The gauge that turned the recursion into $\Phi_{n+1} = L_n(\lambda) \Phi_n$ respects periodicity, since $q_{n+N} = q_n$, so this happens exactly when $T(\lambda)$ has an eigenvalue 1: when $\det(\mathbb{1} - T(\lambda)) = 2 - t(\lambda) = 0$, using $\det T = 1$. For $N \geq 3$, $\det(\lambda - L)$ and $t(\lambda) - 2$ are both monic of degree N . Where the spectrum of L is simple they have the same N roots, so they are equal there. Such points are dense: at fixed q , momenta far enough apart make

the discs of Gershgorin's theorem disjoint and the spectrum simple, so the discriminant of $\det(\lambda - L)$ is a non-zero polynomial in p , and its non-zero set is dense. Both sides are continuous in (q, p) , so *sketched*

$$\det(\lambda - L) = t(\lambda) - 2. \quad (9.6)$$

The c_k are therefore, up to sign and the constant 2, the elementary symmetric functions of the eigenvalues of L . Newton's identities turn them into the $\text{tr } L^k$ and back. *They are Lecture 8's integrals, reorganised*, and they are the candidates for Definition 5.5. Everything now turns on their brackets.

9.3 The notation the proof needs: tensor legs

$L_n(\lambda)$ is a 2×2 matrix whose entries are functions on phase space. What we need is the collection of *all* their brackets, $\{(L_n)_{ab}(\lambda), (L_n)_{cd}(\mu)\}$, sixteen functions at once. The trick is to package them as a single 4×4 matrix acting on $V \otimes V$, with $V = \mathbb{C}^2$.

Definition 9.2 (Tensor legs). Write E_{ab} for the matrix unit with a 1 in row a , column b , and recall $(A \otimes B)_{(ac),(bd)} = A_{ab}B_{cd}$. For 2×2 matrices A, B put $A_1 = A \otimes \mathbb{1}$, $B_2 = \mathbb{1} \otimes B$, and

$$\{A_1, B_2\} = \sum_{a,b,c,d} \{A_{ab}, B_{cd}\} E_{ab} \otimes E_{cd}, \quad \text{that is} \quad \left(\{A_1, B_2\} \right)_{(ac),(bd)} = \{A_{ab}, B_{cd}\}. \quad (9.7)$$

The subscript is a *leg*, not a matrix index: A_1 is the whole matrix A , placed in the first factor (Figure 9.2). The right-hand form of (9.7) is the one to memorise: the entry in row (a, c) and column (b, d) is the bracket of the (a, b) entry with the (c, d) entry.

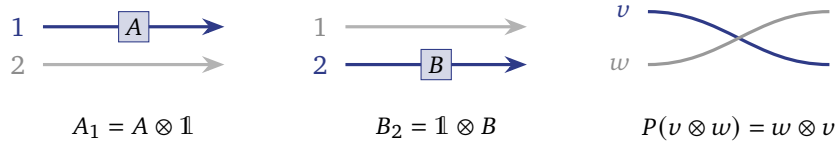


Figure 9.2: Tensor legs. Each horizontal line is a copy of V ; a box on a line is a matrix acting in that factor and doing nothing to the other. Right: the permutation operator is a crossing of the two lines, and $P^2 = \mathbb{1}$ says that uncrossing twice does nothing.

Remark 9.3 (Two brackets in one formula). A_1 and B_2 **commute as matrices**, $(A \otimes \mathbb{1})(\mathbb{1} \otimes B) = A \otimes B = (\mathbb{1} \otimes B)(A \otimes \mathbb{1})$, because they act in different factors. Their **entries do not Poisson-commute**: that is what (9.7) records. The composition theorem below turns on the first fact at its decisive step; nothing uses $\{A_1, B_2\} = 0$, which is false.

One matrix on $V \otimes V$ appears everywhere below.

Definition 9.4 (Permutation operator). $P(v \otimes w) = w \otimes v$; equivalently $P = \sum_{a,b} E_{ab} \otimes E_{ba}$.

Proposition 9.5 (Properties of P). $P^2 = \mathbb{1}$; $P(A \otimes B)P = B \otimes A$, so P commutes with every $A \otimes A$; and

$$(E_{ab} \otimes E_{cd}) P = E_{ad} \otimes E_{cb}. \quad (9.8)$$

Proof. P^2 swaps twice. For (9.8), expand P and use $E_{ab}E_{ef} = \delta_{be}E_{af}$: the sum $\sum_{e,f} \delta_{be}E_{af} \otimes \delta_{df}E_{ce}$ collapses at $e = b, f = d$. The middle property follows by applying both sides to $v \otimes w$. ■

9.4 One identity for one site: the quadratic Sklyanin bracket

Here is the structural assumption that will force involution. It is a statement about the brackets of *one* site matrix at two values of the spectral parameter, and it involves one fixed numerical matrix.

Definition 9.6 (Quadratic Sklyanin bracket). A site matrix $L(\lambda)$ satisfies the *quadratic Sklyanin bracket* with classical r -matrix $r(\lambda)$, a 4×4 matrix of numbers, if

$$\{L_1(\lambda), L_2(\mu)\} = [r(\lambda - \mu), L_1(\lambda) L_2(\mu)] \quad \text{for all } \lambda \neq \mu. \quad (9.9)$$

The left-hand side is sixteen unknown brackets. The right-hand side says they are all determined by the matrices themselves and one numerical r , which is slid past the product $L_1 L_2$ from both sides (Figure 9.3). Why this form and not another is the content of the next theorem: it is the form that survives multiplication.

$$\{L_1, L_2\} = \begin{array}{c} \text{---} \boxed{L} \text{---} \\ \text{---} \boxed{L} \text{---} \end{array} = \begin{array}{c} \text{---} \boxed{L} \text{---} \\ \text{---} \boxed{L} \text{---} \end{array} - \begin{array}{c} \text{---} \boxed{L} \text{---} \\ \text{---} \boxed{L} \text{---} \end{array}$$

Figure 9.3: The quadratic Sklyanin bracket as a picture. Matrices act from right to left along each line. The brackets of the entries of one site matrix with themselves equal the *rung* r acting after the pair $L_1 L_2$, minus the same rung acting before it.

Theorem 9.7 (The quadratic bracket composes). If the chain is ultralocal and every site matrix satisfies (9.9) with the same r , then so does the monodromy matrix: $\{T_1(\lambda), T_2(\mu)\} = [r(\lambda - \mu), T_1(\lambda) T_2(\mu)]$.

Proof. Take $N = 2$; the general case is the same telescoping. Leibniz and ultralocality kill the cross terms, and matrices in different legs commute (Remark 9.3), so with $A = L_1^{(2)} L_2^{(2)}$ and $B = L_1^{(1)} L_2^{(1)}$, the two sites' pairs,

$$\{T_1, T_2\} = [r, A] B + A [r, B] = rAB - ArB + ArB - ABr = [r, AB],$$

and $AB = T_1 T_2$, again because leg-1 and leg-2 matrices commute. ■

Corollary 9.8 (Commuting transfer matrices). $\{t(\lambda), t(\mu)\} = 0$ for all λ, μ . Hence the coefficients c_k of $t(\lambda)$ are in involution.

Proof. Take the trace over both factors, $\text{tr}_{12}(A \otimes B) = \text{tr } A \text{ tr } B$: $\{t(\lambda), t(\mu)\} = \text{tr}_{12} \{T_1, T_2\} = \text{tr}_{12} [r, T_1 T_2] = 0$, because the trace of a commutator vanishes. Expanding in λ and μ , every coefficient Poisson-commutes with every other. At $\lambda = \mu$ the identity follows by continuity. ■

So the whole question of involution has been reduced to one identity for one 2×2 matrix. That is where the work of this lecture is.

9.5 The worked example: the Toda site matrix

Proposition 9.9 (The periodic Toda r -matrix). *Each Toda site matrix (9.2) satisfies the quadratic bracket (9.9) with*

$$r(\lambda) = -\frac{P}{\lambda}. \quad (9.10)$$

Consequently $\{t(\lambda), t(\mu)\} = 0$, and the c_k of (9.5) are in involution.

Proof. Only $L_{11} = \lambda - p$ involves p , so the only non-zero brackets pair it with an off-diagonal entry. With $\{q, p\} = 1$, $\{\lambda - p, e^{-q}\} = -e^{-q}$ and $\{\lambda - p, -e^q\} = -e^q$, both independent of λ and μ . By (9.7),

$$\{L_1(\lambda), L_2(\mu)\} = e^{-q}(E_{12} \otimes E_{11} - E_{11} \otimes E_{12}) - e^q(E_{11} \otimes E_{21} - E_{21} \otimes E_{11}) = (E_{11} \otimes L^0 - L^0 \otimes E_{11})P,$$

with $L^0 = L(0)$; the last step is (9.8) applied to each term. On the right of (9.9), write $L(\lambda) = \lambda E_{11} + L^0$. The permutation commutes with every $A \otimes A$ (Proposition 9.5), which removes the $\lambda\mu$ and the $L^0 \otimes L^0$ terms, and it turns $L^0 \otimes E_{11}$ into $E_{11} \otimes L^0$, so

$$[P, L(\lambda) \otimes L(\mu)] = (\lambda - \mu) [P, E_{11} \otimes L^0] = (\lambda - \mu)(L^0 \otimes E_{11} - E_{11} \otimes L^0)P.$$

Dividing by $-(\lambda - \mu)$ gives the left-hand side. The chain is ultralocal, so Theorem 9.7 lifts the identity to T , and Corollary 9.8 gives the involution. ■

Read the first displayed line as the tensor-leg drill: it is the sixteen brackets of (9.2), of which four are non-zero, written as one 4×4 matrix. *If that line is not comfortable, nothing after it will be.* As an exercise, find the four non-zero brackets and assemble the matrix.

9.6 The theorem

Theorem 9.10 (The periodic Toda chain is Liouville integrable). *On $\mathbb{R}^{2N}$ the coefficients $c_1, \dots, c_N$ of $t(\lambda)$ are in involution and independent on a dense open set, and $H = \frac{1}{2}c_1^2 - c_2$. The periodic Toda chain satisfies Definition 5.5.*

Theorem 9.11 (Tori of the ring · proved modulo Lemma 5.10 and Theorem 1.9). *Let Σ be a connected component of a level set of $(c_1, \dots, c_N)$ on which $dc_1, \dots, dc_N$ are independent at every point, and let $Q = \frac{1}{N} \sum_n q_n$ be the centre of mass. Then $\Sigma \cong T^{N-1} \times \mathbb{R}$, each slice $\Sigma \cap \{Q = \text{const}\}$ is a torus T^{N-1} , and the flow of H is a straight line in the corresponding coordinates.*

9.7 The proofs

Proof of Theorem 9.10. Involution is Proposition 9.9, and (9.5) is $H = \frac{1}{2}c_1^2 - c_2$. So $\{c_k, H\} = c_1 \{c_k, c_1\} - \{c_k, c_2\} = 0$: every c_k is conserved. Conservation comes out of involution and needs no separate argument.

Independence. Read (9.4): $c_k = (-1)^k e_k(p) + R_k$, where e_k is the k -th elementary symmetric polynomial of the momenta and every term of R_k has a factor $e^{q_n - q_{n+1}}$ and degree at most $k-2$ in p (each bond removes two factors $\lambda - p$). In the Jacobian $\det(\partial c_k / \partial p_n)$ the part of top degree in p therefore comes from the e_k alone, and it is $\pm \prod_{i < j} (p_j - p_i)$, the Vandermonde determinant. So at fixed q the Jacobian is a non-zero polynomial in p , which cannot vanish on any open set. Where it does not vanish the dc_k are independent; that set is open, and it meets every open set, so it is dense. ■

The proof of the second theorem has four steps, and only the first two use the Toda chain at all. Write $x_n = e^{q_n - q_{n+1}}$ for the springs.

Proof of Theorem 9.11. (i) Bounded. Every term of H is positive. On Σ the value $H = E$ is fixed, since H is a function of c_1 and c_2 , so $|p_n| \leq \sqrt{2E}$ and $x_n \leq E$ for every n . The springs' exponents telescope to zero round the ring, so $\prod_n x_n = 1$, and each $x_n = \prod_{m \neq n} x_m^{-1} \geq E^{-(N-1)}$. (By the inequality of arithmetic and geometric means $E \geq \sum_n x_n \geq N$, so these bounds are not vacuous.) Every gap $q_n - q_{n+1}$ therefore lies in $[-(N-1) \ln E, \ln E]$. Each $q_n - Q$ is a fixed linear combination of the gaps, so at fixed Q every q_n is bounded too. The slice $\Sigma \cap \{Q = \text{const}\}$ is closed and bounded: compact.

(ii) Complete. Let $G = \sum_k t_k c_k$ for constants t_k . By involution H is constant along the flow of G , so the bounds of (i) hold along every integral curve. The components of X_G are polynomials in the p_n and the x_n , hence bounded by a constant C on the curve, which in a time τ moves at most a distance $C\tau$ and so stays in a compact ball. By the escape argument of Proposition 1.7, a curve that exists only up to a finite time must leave every compact set. So every such flow is complete.

(iii) Cylinder. Stages (i)–(iii) of the proof of Theorem 5.7 and the first half of stage (iv) (§ 5.5) use involution, independence on Σ , connectedness and complete flows, and nothing else. They make Σ the orbit of an action of $\mathbb{R}^N$ with a discrete stabiliser Λ , so $\Sigma \cong \mathbb{R}^N / \Lambda$ (Lemma 5.10).

(iv) Drift. Shifting every p_n by s is the same as shifting λ by $-s$ in (9.4), so $\sum_n \partial c_k / \partial p_n$ is a constant multiple of c_{k-1} , constant on Σ . Hence Q changes at a constant rate along each flow, and Q at the time- s point is $Q(x_0) + \ell(s)$ for a linear map $\ell : \mathbb{R}^N \rightarrow \mathbb{R}$, not zero since $\ell(e_1) = -1$. A period does not move Q , so $\Lambda \subset \ker \ell \cong \mathbb{R}^{N-1}$, and $\Sigma \cap \{Q = Q(x_0)\} \cong \ker \ell / \Lambda$. That is compact by (i), so Λ is a lattice of full rank in $\ker \ell$ (Lemma 5.10): the slice is T^{N-1} and $\Sigma \cong T^{N-1} \times \mathbb{R}$. Stage (vi) of the proof of Theorem 5.7 makes the flow of H linear. ■

Step (i) is the only place the *ring* matters. On the open chain nothing ties the springs together, a single spring can be made as weak as one likes at fixed energy, and the level sets stay unbounded even after the drift is removed: the open chain scatters instead of winding.

9.8 What it buys

The motion. On every regular level set the flow of H is a straight line in coordinates on $T^{N-1} \times \mathbb{R}$: it winds round the torus while the centre of mass moves along $\mathbb{R}$ at the constant speed $\sum_n p_n / N$, possibly zero. For $N = 3$, with the centre of mass removed, the tori meet the Poincaré section of Figure 7.5 in its closed curves: the numerical plot of Lecture 7, beside the chaos of Hénon–Heiles at the same energy, illustrates this theorem at $E = 1/6$.

Action–angle variables need one more step. Separate the centre of mass canonically, fix the total momentum, and $N - 1$ relative degrees of freedom remain, whose regular level sets are the compact tori of Theorem 9.11. To those Theorem 6.5 applies. As an exercise, carry out the separation for $N = 2$ with $Q = (q_1 + q_2)/2$, $2\bar{p} = p_1 + p_2$, $x = q_1 - q_2$ and $p_x = (p_1 - p_2)/2$; the two-particle ring at the end of the readings does it. Computing the actions needs the curve $z + 1/z = t(\lambda)$ of the reading below, with one action for each of its $N - 1$ gaps · cited: Flaschka and McLaughlin 1976.

The method. Three moves did everything, and none of them is special to Toda:

1. one identity, (9.9), for *one* site matrix;
2. composition along the chain, Theorem 9.7, which needs only ultralocality;

3. the trace, which kills a commutator.

Change the site matrix and the r -matrix, and the same three moves give commuting integrals for another chain. [Lecture 14](#) replaces the Poisson bracket by the commutator and r by an R -matrix with $R = \mathbb{1} + i\hbar r + O(\hbar^2)$. The same three moves then give the commuting transfer matrices of the Heisenberg spin chain, and on the quantum Toda chain their limit $\hbar \rightarrow 0$ is exactly (9.9) with (9.10). That is goal G5, and the one row of the classical-to-quantum dictionary that is a theorem rather than an analogy.

9.9 Caveats, in one place

Three things a careful reader should know, collected here so that the arguments above can run without interruption. (The linear bracket, which this lecture does not need, is in the reading.)

1. *Independence is generic, not universal.* [Theorem 9.10](#) proves independence on a dense open set, and that is all [Definition 5.5](#) asks. It does fail somewhere: at the equilibrium $p_n = 0$, $x_n = 1$ every flow is a drift.
2. *The ring drifts.* The flow of $c_1 = -\sum_n p_n$ translates every q_n and changes no spring, so no level set is compact. Tori appear only after the centre of mass is fixed, which is why [Theorem 9.11](#) is stated for slices.
3. *The sign of r belongs to the site matrix.* Other books use the site matrix with $e^{\pm q}$ exchanged, which gives $r = +P/\lambda$; mixing a sign with the wrong site matrix breaks (9.9).

Reading

9.10 The same integrals from zero curvature, and the route not taken

Conservation directly. ([Lecture 8](#)'s reading has the fuller version, with the plaquette picture.) Involution gave conservation for free, but $t(\lambda)$ is also conserved for a more geometric reason. With $V_n(\lambda) = \begin{pmatrix} -\lambda/2 & -e^{-q_n} \\ e^{q_{n-1}} & \lambda/2 \end{pmatrix}$, the equations of motion of the ring are equivalent to $\dot{L}_n = V_{n+1}L_n - L_nV_n$ for every n and every λ : transport in space and in time commute round every plaquette of the space-time lattice, a discrete *zero-curvature* condition. Differentiating the product (9.3), the sum telescopes to $\dot{T} = V_{N+1}T - TV_1 = [V_1, T]$, a Lax equation for the 2×2 matrix T , and $t = \text{tr } T$ is conserved. Periodicity is used once, to identify V_{N+1} with V_1 .

The route not taken. One can prove integrability of the *open* chain instead: its $N \times N$ Lax matrix satisfies the linear bracket with a constant r , the involution theorem for that bracket gives $\{\text{tr } L^j, \text{tr } L^k\} = 0$, and independence follows from a band lemma for tridiagonal matrices. That route reaches the same kind of theorem for a different chain, and §§ 9.11 and 9.12 take it. Its bracket does not compose, so it cannot be quantised site by site, and it is the ring's quadratic bracket that [Lecture 14](#) quantises.

The spectral curve. Setting $\det(z - T(\lambda)) = 0$ gives $z + 1/z = t(\lambda)$, since $\det T = 1$. Its $N - 1$ *finite gaps* are the bounded intervals of real λ where $|t(\lambda)| > 2$, and when all are open they carry the $N - 1$ action variables. § 9.13 builds the curve from Bloch waves and proves these statements, or cites them.

9.11 The route not taken, I: the linear bracket

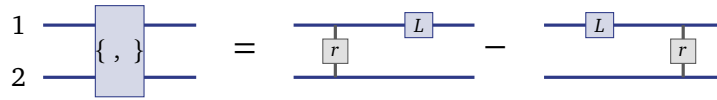
The core proved the ring's integrals commute through the *quadratic* bracket, because it composes. The older route goes through a bracket *linear* in L , for a single Lax matrix; the open-chain reading below applies it.

Definition 9.12 (Sklyanin bracket). A Lax matrix $L(\lambda)$ satisfies a *Sklyanin bracket* with classical r -matrix r if

$$\{L_1(\lambda), L_2(\mu)\} = [r_{12}(\lambda - \mu), L_1(\lambda) + L_2(\mu)], \quad (9.11)$$

where $r_{12}(\lambda)$ is a matrix on $V \otimes V$ of *numbers* — it does not depend on the point of phase space.

Read what (9.11) asserts. On the left is an arbitrary $d^2 \times d^2$ matrix of Poisson brackets, d^4 functions on phase space with no reason to be related. On the right is a commutator of one fixed numerical matrix with something built linearly from L itself. The assumption is enormously restrictive, and its whole force is that *the same r works at every point of phase space and for every pair λ, μ .*



r is a *rung* joining the two legs; the commutator is the difference of the two orders.

Figure 9.4: The Sklyanin bracket (9.11), drawn for the L_1 half of $L_1 + L_2$; the L_2 half is the same picture with the box on the lower leg. The left-hand side is an unknown matrix of Poisson brackets; the right-hand side says it equals one fixed rung r slid past L , minus L slid past the rung. Note that the two terms are added, not multiplied — the picture with an L on *both* legs is the *quadratic* bracket of § 9.4, which is a different relation.

Two qualifications:

- *Numerical.* If r is allowed to depend on the dynamical variables the bracket is called *dynamical*, and optional lecture B meets the first real example — Calogero–Moser has no r of the constant form. Everything proved in the involution theorem below survives that generalisation; the classical Yang–Baxter equation of [Lecture 14](#)'s reading does not.
- *Which r ?* Part I teaches how to *read and check* an r -matrix, never how to solve for one. Finding r from scratch is outside it.

Remark 9.13 (r is not arbitrary). Antisymmetry of the Poisson bracket already constrains r . Reading (9.7) backwards, $\{L_{ab}, L_{cd}\} = -\{L_{cd}, L_{ab}\}$ is the matrix statement $\{L_1(\lambda), L_2(\mu)\} = -P \{L_1(\mu), L_2(\lambda)\} P$. Substituting (9.11) on both sides and writing $r_{21} = P r_{12} P$, the two agree as soon as

$$r_{12}(\lambda) = -r_{21}(-\lambda). \quad (9.12)$$

The rational $r(\lambda) = cP/\lambda$ satisfies it, since $PPP = P$ makes $r_{21} = r_{12}$ there, and cP/λ is odd. The classical Yang–Baxter equation of [Lecture 14](#)’s reading uses (9.12) to trade r_{21} for $-r_{12}$.

Reading

The involution theorem

Theorem 9.14 (Involution from the Sklyanin bracket). *If $L(\lambda)$ satisfies (9.11), then for all $n, m \geq 1$ and all λ, μ at which it holds,*

$$\{\mathrm{tr} L(\lambda)^n, \mathrm{tr} L(\mu)^m\} = 0. \quad (9.13)$$

The qualification is needed because the r -matrices that occur have a pole at coincident arguments — the rational one is $cP/(\lambda - \mu)$ — so (9.11) is an assumption about $\lambda \neq \mu$ and says nothing at $\lambda = \mu$. Nothing is lost: the left of (9.13) is continuous in λ and μ whenever L is, and a continuous function vanishing off the diagonal vanishes on it too.

The proof is three steps and each is short.

Proof. Step 1: the chain rule, in tensor-leg form. Both $\mathrm{tr} L(\lambda)^n$ and $\mathrm{tr} L(\mu)^m$ are functions of the entries of L , so by bilinearity and the Leibniz rule of [Lecture 2](#),

$$\{F, G\} = \sum_{a,b,c,d} \frac{\partial F}{\partial L_{ab}} \frac{\partial G}{\partial L_{cd}} \{L_{ab}, L_{cd}\}.$$

For $F = \mathrm{tr} L^n$ we have $\partial F / \partial L_{ab} = n (L^{n-1})_{ba}$, and similarly for G . Comparing with (9.7), the double sum is exactly a trace over both legs:

$$\{\mathrm{tr} L(\lambda)^n, \mathrm{tr} L(\mu)^m\} = nm \, \mathrm{tr}_{12} \left(L_1(\lambda)^{n-1} L_2(\mu)^{m-1} \{L_1(\lambda), L_2(\mu)\} \right). \quad (9.14)$$

Step 2: substitute the assumption. Put $A = L_1^{n-1} L_2^{m-1}$ and $B = L_1 + L_2$. Then (9.11) turns (9.14) into

$$\{\mathrm{tr} L(\lambda)^n, \mathrm{tr} L(\mu)^m\} = nm \, \mathrm{tr} (A [r, B]).$$

Step 3: the trace of that vanishes. Here is the only place any structure is used, and it is [Remark 9.3](#): L_1 and L_2 commute as matrices, so A — a product of their powers — commutes with B . Hence, using cyclicity of the trace twice and $AB = BA$ once,

$$\mathrm{tr} (A [r, B]) = \mathrm{tr}(ArB) - \mathrm{tr}(ABr) = \mathrm{tr}(ArB) - \mathrm{tr}(rAB) = \mathrm{tr}(ArB) - \mathrm{tr}(rBA) = \mathrm{tr}(ArB) - \mathrm{tr}(ArB) = 0. \quad \blacksquare$$

This pays [Lecture 8](#)’s debt by the route not taken. Note how little was spent: the argument never opened r , never used the Lax equation, and never mentioned a Hamiltonian. It also never used the *particular* combination $L_1 + L_2$ — only that it commutes with A .

Corollary 9.15 (What the model gets). *Expanding $\mathrm{tr} L(\lambda)^k$ in powers of λ produces a family of functions in involution, and any Hamiltonian among them has all the others as commuting integrals. Whether there are n independent ones is a separate question about the model, settled for the open Toda chain in the open-chain reading below.*

The converse: Babelon–Viallet

Theorem 9.14 is one-way: a Sklyanin bracket gives involution. The converse is true but weaker than one would like.

Theorem 9.16 (Babelon–Viallet · cited). *If the eigenvalues of L are in involution on a neighbourhood where L is regular and semisimple, then L satisfies*

$$\{L_1, L_2\} = [r_{12}, L_1] - [r_{21}, L_2], \quad (9.15)$$

with r in general dynamical — a function on phase space.

Read (9.15) carefully: it is not (9.11). Two independent matrices appear, and the single-commutator form is the special case $r_{21} = -r_{12}$. The converse does *not* deliver that special case, and the difference is not a technicality — it is the difference between a true statement and a false one.

Example 9.17 (Why the strengthening fails). On a canonical (q, p) plane put $D = \text{diag}(a, b)$ with $a \neq b$, and

$$L(q, p) = UDU^{-1}, \quad U = e^{qE_{12}}e^{pE_{21}}.$$

Then L is regular semisimple near the origin and its eigenvalues are the *constants* a, b , so they are in involution for the most trivial of reasons. At the origin $\partial_q L = [E_{12}, D] = (b - a)E_{12}$ and $\partial_p L = [E_{21}, D] = (a - b)E_{21}$, so

$$\{L_{12}, L_{21}\} \Big|_0 = -(a - b)^2 \neq 0.$$

Now look at where that number would have to come from in (9.11). At the origin $L_1 + L_2 = D \otimes \mathbb{1} + \mathbb{1} \otimes D$ is diagonal with entries $2a, a + b, a + b, 2b$, and the required entry sits between the two *equal* ones. For any matrix r whatever, dynamical or not, a commutator with a diagonal matrix has that entry multiplied by the difference of the two diagonal entries — here zero. So no r can satisfy (9.11) for this L , while (9.15), with its two independent terms, has no such obstruction.

So an r -matrix always exists behind involution, but it need not be numerical and it need not put the bracket in the single-commutator form of Definition 9.12. The classification of solutions of the classical Yang–Baxter equation (Lecture 14’s reading) applies to neither. Optional lecture B meets a dynamical r in Calogero–Moser, so the caveat is not hypothetical.

9.12 The route not taken, II: the open chain, proved directly

The open Toda chain is the ring with its closing spring removed: $H = \frac{1}{2} \sum_{n=1}^N p_n^2 + \sum_{n=1}^{N-1} e^{q_n - q_{n+1}}$, the chain whose scattering Lecture 12 solves. This reading proves it Liouville integrable with the linear bracket, and shows what the torus theorem does and does not give when every fibre is non-compact.

The phase space is $\mathbb{R}^{2N}$ with coordinates (q, p) , so $n = N$ degrees of freedom. § 8.4 put the equations of motion in Lax form, with Flaschka’s variables (8.4), $a_n = e^{(q_n - q_{n+1})/2}$ and

$b_n = p_n$, and

$$L = \begin{pmatrix} b_1 & a_1 & & \\ a_1 & b_2 & \ddots & \\ & \ddots & \ddots & a_{N-1} \\ & & a_{N-1} & b_N \end{pmatrix}, \quad M_{n,n+1} = -\frac{1}{2}a_n = -M_{n+1,n}, \quad \dot{L} = [M, L] \quad (9.16)$$

(**Proposition 8.3**). By **Theorem 8.2** the eigenvalues of L are conserved, and so are the N functions

$$F_k = \text{tr } L^k, \quad k = 1, \dots, N, \quad (9.17)$$

which begin $F_1 = \sum_n p_n$, $F_2 = 2H$ (8.8). They are our candidates for **Definition 5.5**, with $H = \frac{1}{2}F_2$.

Three facts about L will be used over and over.

- (L1) *Simple spectrum*. Every $a_n > 0$, so L is a real symmetric (Jacobi) matrix with N distinct real eigenvalues $\lambda_1 < \dots < \lambda_N$ (proof of **Proposition 8.3**). Here λ_k are eigenvalues, not a spectral parameter: the open chain has none.
- (L2) *Bounded entries*. For a real symmetric matrix $|L_{ij}| \leq \max_k |\lambda_k|$, since $|L_{ij}| = |e_i \cdot L e_j|$ is at most the largest stretching factor of L . The eigenvalues are conserved, so along any motion every entry of L stays bounded by a constant fixed at $t = 0$.
- (L3) *Brackets*. Since $\{f, p_m\} = \partial f / \partial q_m$,

$$\{a_n, b_m\} = \frac{1}{2}a_n (\delta_{nm} - \delta_{m,n+1}), \quad \{a_n, a_m\} = \{b_n, b_m\} = 0. \quad (9.18)$$

The bond variable a_n is shifted only by the momenta at its two ends, with opposite signs.

Reading

Independence, by a Vandermonde argument

Three particles, by hand. For $N = 3$ the traces are $F_1 = \sum p_n$, $F_2 = \sum p_n^2 + 2(a_1^2 + a_2^2)$ and $F_3 = \sum p_n^3 + 3a_1^2(p_1 + p_2) + 3a_2^2(p_2 + p_3)$. Their derivatives with respect to the momenta alone form a 3×3 matrix with determinant

$$\det \frac{\partial(F_1, F_2, F_3)}{\partial(p_1, p_2, p_3)} = 6 \left[(p_2 - p_1)(p_3 - p_1)(p_3 - p_2) - a_1^2(p_2 - p_1) - a_2^2(p_3 - p_2) \right]. \quad (9.19)$$

When the particles are far apart, or their momentum *differences* large compared with the a_n (a common boost changes nothing), the a_n^2 terms are negligible, the F_k are nearly the power sums $\sum p_n^k$, and (9.19) is six times the *Vandermonde determinant* of the momenta: non-zero as soon as the momenta are distinct. That is why the far-apart (or large momentum-separation) regime is the right place to look first: there the chain is three free particles, and the three integrals are visibly three different functions.

But (9.19) is not a proof of anything everywhere. It vanishes when all three momenta are equal, and also at $p = (0, 1, 2)$, $a_1 = a_2 = 1$, where the momenta are distinct. At such points independence has to come from the q -derivatives. The general proof uses both, and is organised by L .

General N .

Lemma 9.18 (Gradients of the traces). *With the conventions $a_0 = a_N = 0$,*

$$\frac{\partial F_k}{\partial p_n} = k (L^{k-1})_{nn}, \quad \frac{\partial F_k}{\partial q_n} = k \left[a_n (L^{k-1})_{n,n+1} - a_{n-1} (L^{k-1})_{n-1,n} \right]. \quad (9.20)$$

Proof. For any variable x , $\partial_x \operatorname{tr} L^k = k \operatorname{tr} (L^{k-1} \partial_x L)$, by the product rule and cyclicity (as in Theorem 8.2(i)). Now $\partial L / \partial p_n$ is the matrix unit E_{nn} , and a_n depends on q_n and q_{n+1} with $\partial a_n / \partial q_n = \frac{1}{2} a_n$, $\partial a_n / \partial q_{n+1} = -\frac{1}{2} a_n$. Each a_m sits in two symmetric slots and L^{k-1} is symmetric, which turns the two halves into one. ■

Lemma 9.19 (The band lemma). *Let L be a Jacobi matrix with all $a_n \neq 0$, and let X be a symmetric matrix commuting with L whose diagonal and first off-diagonals vanish. Then $X = 0$.*

Proof. Suppose not, and let $m \geq 2$ be the smallest distance from the diagonal at which X has a non-zero entry; write $x_i = X_{i,i+m}$. Compute the entry $(i, i+m-1)$ of $LX - XL$. Every entry of X closer than m to the diagonal is zero, and L is tridiagonal, so only one term survives from each product:

$$0 = [L, X]_{i,i+m-1} = a_{i-1} x_{i-1} - a_{i+m-1} x_i.$$

At $i = 1$ the first term is absent, and $a_m \neq 0$ gives $x_1 = 0$; then $i = 2$ gives $x_2 = 0$, and so on along the whole diagonal. So the m -th diagonal vanishes after all, a contradiction. ■

Both conditions on X are needed: $X = \mathbb{1}$ commutes with L and has a zero off-diagonal; and for $N = 2$ with $b_1 = b_2$, $X = E_{12} + E_{21}$ commutes with L and has a zero diagonal.

Theorem 9.20 (Independence). *The integrals $F_1, \dots, F_N$ of (9.17) are independent at every point of $\mathbb{R}^{2N}$.*

Proof. Suppose $\sum_k c_k dF_k = 0$ at some point. Put $g(\xi) = \sum_{k=1}^N k c_k \xi^{k-1}$, a polynomial of degree at most $N-1$, and $X = g(L)$, a symmetric matrix commuting with L . By (9.20), the p_n -component of $\sum_k c_k dF_k$ is X_{nn} and the q_n -component is $a_n X_{n,n+1} - a_{n-1} X_{n-1,n}$. All vanish. The first set says the diagonal of X is zero. The second, read at $n = 1, 2, \dots$ in turn, says $a_1 X_{12} = 0$, then $a_2 X_{23} = a_1 X_{12} = 0$, and so on: the first off-diagonal is zero. By Lemma 9.19, $X = 0$.

So $g(\lambda_j) = 0$ at each of the N distinct eigenvalues (L_1) . In the coefficients c_k this is N linear equations whose matrix $(k \lambda_j^{k-1})$ has determinant $N! \prod_{i < j} (\lambda_j - \lambda_i) \neq 0$, the Vandermonde determinant again. Hence every $c_k = 0$. ■

The Vandermonde determinant of the $N = 3$ example reappears at the end, with the momenta replaced by the eigenvalues, which are distinct at every point. That upgrades “independent where the momenta are distinct” to “independent everywhere”. Definition 5.1 asks only for a dense open set; § 9.12 needs the stronger statement.

Reading

Involution, from the open chain's r -matrix

Theorem 9.14 gives $\{\operatorname{tr} L^j, \operatorname{tr} L^k\} = 0$ as soon as L satisfies a Sklyanin bracket (9.11). The open chain's L has no spectral parameter, so we look for a constant matrix r on $\mathbb{R}^N \otimes \mathbb{R}^N$ with

$$\{L_1, L_2\} = [r, L_1 + L_2]. \quad (9.21)$$

The left-hand side. By (9.7) the entry of $\{L_1, L_2\}$ in row (a, c) and column (b, d) is $\{L_{ab}, L_{cd}\}$. By (9.18) it is non-zero only when one of the two entries is a diagonal b_m and the other is an a_n in slot $(n, n+1)$ or $(n+1, n)$.

An exercise before the proof. For $N = 2$, with $a = a_1$, the sixteen entries collapse to

$$\{L_1, L_2\} = \frac{a}{2} \left[(E_{12} + E_{21}) \otimes (E_{11} - E_{22}) - (E_{11} - E_{22}) \otimes (E_{12} + E_{21}) \right] :$$

eight non-zero entries, each $\pm a/2$. Write out the 4×4 matrix in § 9.3's ordering and compare it, entry by entry, with $[r, L_1 + L_2]$ for $r = \frac{1}{2}(E_{21} \otimes E_{12} - E_{12} \otimes E_{21})$ before reading the general proof.

The right-hand side, and the match.

Proposition 9.21 (The open Toda r -matrix). *The constant matrix*

$$r = \frac{1}{2} \sum_{i < j} (E_{ji} \otimes E_{ij} - E_{ij} \otimes E_{ji}) \quad (9.22)$$

satisfies (9.21) for the Toda L . Consequently $\{F_j, F_k\} = 0$ for all j, k .

Proof. Put $\epsilon_{ij} = +1$ if $i < j$, -1 if $i > j$, 0 if $i = j$. Then $r = \frac{1}{2} \sum_{i,j} \epsilon_{ij} E_{ji} \otimes E_{ij}$, and

$$[r, L_1 + L_2] = \frac{1}{2} \sum_{i,j} \epsilon_{ij} \left([E_{ji}, L] \otimes E_{ij} + E_{ji} \otimes [E_{ij}, L] \right).$$

Expanding $[E_{ji}, L] = \sum_k (L_{ik} E_{jk} - L_{kj} E_{ki})$ and collecting the coefficient of $E_{ab} \otimes E_{cd}$ gives, for any matrix L ,

$$[r, L_1 + L_2]_{(ac),(bd)} = \frac{1}{2} (\epsilon_{ab} + \epsilon_{cd}) (L_{cb} \delta_{ad} - L_{ad} \delta_{bc}). \quad (9.23)$$

Compare with $\{L_{ab}, L_{cd}\}$, case by case.

- $c = d$, a diagonal entry. Then $\epsilon_{cd} = 0$ and (9.23) is $\frac{1}{2} \epsilon_{ab} L_{ab} (\delta_{ac} - \delta_{bc})$. For $(a, b) = (n, n+1)$ this is $\frac{1}{2} a_n (\delta_{nc} - \delta_{n+1,c}) = \{a_n, b_c\}$; for $(n+1, n)$ both signs flip and it is the same. For $|a - b| \geq 2$, $L_{ab} = 0$, as is the bracket; for $a = b$, both are zero.
- $a = b$. The same computation with the roles exchanged gives $-\{L_{cd}, b_a\} = \{b_a, L_{cd}\}$.
- Both off-diagonal. The bracket is zero. In (9.23) the prefactor vanishes unless both slots lie above the diagonal or both below. Say $a < b$ and $c < d$. If $a = d$ then $c < a < b$, so $|c - b| \geq 2$ and $L_{cb} = 0$; if $b = c$ then $a < b < d$ and $L_{ad} = 0$. Below the diagonal is the same.

So (9.21) holds entry by entry, and Theorem 9.14 gives the involution. ■

The model entered only in the last case: (9.23) holds for every matrix, and it matches the brackets because L is *tridiagonal*. (scripts/check-19-algebra.py verifies (9.23) and (9.21) symbolically for $N \leq 5$.)

The hypothesis audit: what we have, and what we have not

Theorem 9.22 (The open Toda chain is Liouville integrable). *On $\mathbb{R}^{2N}$, the functions $F_k = \text{tr } L^k$, $k = 1, \dots, N$, are independent at every point and in involution, and $H = \frac{1}{2}F_2$. The open Toda chain satisfies Definition 5.5.*

Proof. Theorem 9.20, Proposition 9.21, and (8.8). ■

That is the definition. Now the theorem. Theorem 5.7 is about a connected component Σ of a level set $\mathcal{M}_f = \{F = f\}$. Its hypotheses, and the places its proof spends them:

hypothesis	open Toda	where § 5.5 spends it
involution	✓ Proposition 9.21	stages (i)–(iii)
(I) independence on Σ	✓ Theorem 9.20, everywhere	stages (i), (iii), (iv)
Σ connected	✓ by choice of Σ	stage (iii)
(C) Σ compact	✗ fails for every Σ	stage (ii) completeness; stage (iv) full rank

Compactness fails. The flow of $F_1 = \sum_n p_n$ is $q_n \mapsto q_n + s$, $p_n \mapsto p_n$: a rigid translation of the whole chain. It changes no a_n and no b_n , so it preserves every F_k and maps Σ into itself, and its orbits are unbounded lines. A compact set contains no unbounded line. No level set of the open chain is compact, and Theorem 5.7 does not apply to any of them.

That does not end the audit. § 5.5 used compactness in two places, and one of them can be repaired by hand.

Proposition 9.23 (Completeness). *For any constants $t_1, \dots, t_N$, the Hamiltonian vector field of $G = \sum_k t_k F_k$ is complete. In particular the flows of $F_1, \dots, F_N$ are.*

Proof. The components of X_G are $\partial G / \partial p_n$ and $-\partial G / \partial q_n$, which by (9.20) are polynomials in the entries of L . Along the flow of G each F_k is conserved, since $\{F_k, G\} = \sum_j t_j \{F_k, F_j\} = 0$; the F_k fix the characteristic polynomial of L , so its spectrum is constant along an integral curve, and by (L2) the entries of L are bounded on it by a constant. Hence $|X_G| \leq C$ along the curve, which in a time T moves at most a distance CT and so stays in a compact ball. By the escape argument in the proof of Proposition 1.7, a curve that exists only up to a finite time must leave every compact set. So the curve exists for all time. ■

With completeness in hand, stages (i)–(iii) of § 5.5 and the first half of stage (iv) go through word for word: they use involution, (I), connectedness, and complete flows, and nothing else.

Corollary 9.24 (What survives · proved modulo Lemma 5.10 and Theorem 1.9). *Every connected component Σ of every level set of $(F_1, \dots, F_N)$ is a Lagrangian submanifold of $\mathbb{R}^{2N}$ diffeomorphic to $T^k \times \mathbb{R}^{N-k}$ for some $k < N$, and the flows of the F_k are straight lines in the corresponding coordinates.*

Proof. Stage (iv) with a discrete stabiliser gives $T^k \times \mathbb{R}^{N-k}$ for some $0 \leq k \leq N$ (Lemma 5.10); $k = N$ would make Σ compact. ■

The torus conclusion of **Theorem 5.7** is not claimed for the open chain, and it is false. **Lecture 12**'s reading shows that $k = 0$: every fibre is a copy of $\mathbb{R}^N$, with no periodic direction at all. If this feels like a let-down after so many tori, it is the point. The definition of integrability is algebraic (**Definition 5.5**); the torus is a *theorem*, with a hypothesis, and here the hypothesis fails. What the open chain does on its non-compact fibres is the subject of **Lecture 12**: it scatters.

Reading

9.13 Bloch–Floquet and the spectral curve

The core's transfer matrix is the 2×2 face of the ring's integrals. This reading finds the $N \times N$ face, a Lax matrix $L(z)$ with a second spectral parameter, and shows the two define one curve, whose gaps carry the actions.

The eigenvalue problem $L\psi = \lambda\psi$ of the ring is the recursion (8.13) of **Lecture 8**, closed up by $\psi_0 = \psi_N$ and $\psi_{N+1} = \psi_1$. Now unroll the ring (**Figure 9.5**): extend a_n, b_n periodically and let (8.13) hold for every $n \in \mathbb{Z}$, an electron on a crystal whose unit cell is our ring. The ring's eigenvectors are the solutions of period N , but the crystal has more states, and Bloch's theorem (from solid-state physics) says which: the bounded solutions can be chosen with

$$\psi_{n+N} = z\psi_n \quad \text{for all } n, \quad z = e^{i\theta}, \quad (9.24)$$

θ the Bloch phase per unit cell. We impose (9.24) for every non-zero complex z , the *Bloch–Floquet multiplier*.

A solution of (9.24) is fixed by $\psi_1, \dots, \psi_N$. Rows 2 to $N-1$ of (8.13) are rows of L and are unchanged. Row 1 contains $a_0\psi_0 = a_N\psi_0 = a_N z^{-1}\psi_N$, and row N contains $a_N\psi_{N+1} = a_N z\psi_1$. So the Bloch condition turns (8.13) into $L(z)\psi = \lambda\psi$ with

$$L(z) = \begin{pmatrix} b_1 & a_1 & & & a_N/z \\ a_1 & b_2 & a_2 & & \\ & \ddots & \ddots & \ddots & \\ & & a_{N-2} & b_{N-1} & a_{N-1} \\ a_N z & & & a_{N-1} & b_N \end{pmatrix}, \quad L(1) = L. \quad (9.25)$$

Nothing was postulated: $L(z)$ is the crystal's Hamiltonian restricted to Bloch waves.

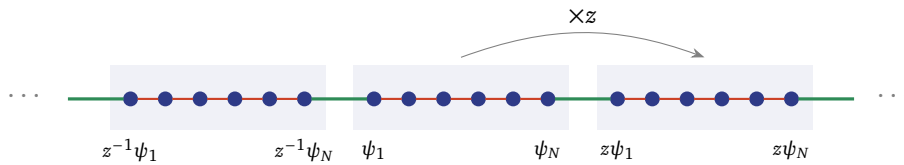


Figure 9.5: The ring unrolled into a crystal whose unit cell (shaded) is the ring. A Bloch wave repeats from cell to cell up to the factor z . Only the bonds that cross a cell boundary (green) see z , hence the corners of $L(z)$. The ring itself is $z = 1$.

Proposition 9.25 (A Lax pair for every z). Let $M(z)$ agree with M except in the corners, $M(z)_{N1} = -\frac{1}{2}a_N z$ and $M(z)_{1N} = \frac{1}{2z}a_N$. Then along every solution of (8.10), $\dot{L}(z) = [M(z), L(z)]$ for every $z \neq 0$. In particular the spectrum of $L(z)$, and $\det(\lambda - L(z))$ as a

function of two variables, are conserved.

The proof restricts the infinite crystal's Lax equation to Bloch waves.

Proof of Proposition 9.25. Let L_∞ and M_∞ be the infinite periodic tridiagonal matrices acting on sequences $(\psi_n)_{n \in \mathbb{Z}}$, with the entries of L and M in every cell. Entry by entry, $\dot{L}_\infty = [M_\infty, L_\infty]$ is the open-chain computation of Proposition 8.3 with no ends at all, so it holds. Both matrices commute with the shift $\psi_n \mapsto \psi_{n+N}$, since their entries are periodic, so both map the space of solutions of (9.24) into itself. On that N -dimensional space, in the coordinates $\psi_1, \dots, \psi_N$, the derivation of (9.25) says L_∞ acts as $L(z)$, and the same derivation applied to M_∞ gives $M(z)$. Restricting the identity to the invariant subspace gives the result. ■

At $z = e^{i\theta}$, $L(z)$ is Hermitian and its eigenvalues are the Bloch energies at phase θ ; as θ runs round the circle they sweep out the *bands* of the crystal (below), which by Proposition 9.25 do not change as the particles move.

A word on names. In § 8.11 the matrix carried λ and its eigenvalue was η . Here the matrix carries z and its eigenvalue is λ . That is deliberate: the eigenvalue of $L(z)$ is the spectral parameter of the core's 2×2 site matrix (spectral duality, below).

Reading

The spectral curve, bands and gaps

Now put the two constructions side by side: the $N \times N$ matrix $L(z)$ of § 9.13 and the 2×2 matrix $T(\lambda)$ of § 9.2. They came from the same recursion.

Proposition 9.26 (Bloch waves are eigenvectors of the monodromy matrix). *For $z \neq 0$ and any λ , the map $\psi \mapsto \Phi_1$ is a linear isomorphism from the eigenspace of $L(z)$ for the eigenvalue λ onto the eigenspace of $T(\lambda)$ for the eigenvalue z . In particular λ is an eigenvalue of $L(z)$ if and only if z is an eigenvalue of $T(\lambda)$, and then*

$$z + \frac{1}{z} = t(\lambda). \quad (9.26)$$

Proof. Given $L(z)\psi = \lambda\psi$, extend ψ to all n by (9.24); it solves (8.13) for every n . Gauge to $\phi_n = e^{-q_n/2}\psi_n$ and form Φ_n . By (8.15), $\Phi_{N+1} = T(\lambda)\Phi_1$, and by periodicity of the q_n , $\Phi_{N+1} = z\Phi_1$. If $\Phi_1 = 0$ then $\phi_0 = \phi_1 = 0$, and (8.14) run both ways gives $\phi \equiv 0$; so the map is injective. Conversely, an eigenvector Φ_1 of $T(\lambda)$ generates $\Phi_n = L_{n-1} \cdots L_1 \Phi_1$ (and, using L_n^{-1} , which exists since $\det L_n = 1$, for $n < 1$), whose first components solve (8.14), with $\Phi_{n+N} = z\Phi_n$ because the site matrices are periodic; ungauging gives a Bloch solution of (8.13), which is an eigenvector of $L(z)$ by § 9.13. Finally $\det T = 1$, so the eigenvalues of T are z and $1/z$, and their sum is $t(\lambda)$. ■

Definition 9.27 (Spectral curve). The *spectral curve* of the periodic Toda chain is $\Gamma = \{(\lambda, z) \in \mathbb{C} \times \mathbb{C}^\times : z + 1/z = t(\lambda)\}$.

The curve does not move, since every c_k is conserved (proof of Theorem 9.10). Its coefficients are the integrals, and it holds all of them at once. Spectral curves of this kind, and their quantum counterparts, return in the reading of optional lecture B as a frontier.

Bands and gaps. Take λ real. Then $t(\lambda)$ is real and the two roots of $z^2 - t(\lambda)z + 1 = 0$ are

$$z_{\pm} = \frac{t(\lambda) \pm \sqrt{t(\lambda)^2 - 4}}{2}, \quad z_+ z_- = 1.$$

- If $|t(\lambda)| \leq 2$, they are $e^{\pm i\theta}$ with $2 \cos \theta = t(\lambda)$: the recursion has bounded Bloch solutions, the electron propagates, and λ is in a *band*.
- If $|t(\lambda)| > 2$, they are real, one of modulus above 1 and one below: every non-zero solution is unbounded along the crystal, and λ is in a *gap*.
- The *band edges* are where $t(\lambda) = \pm 2$, that is $z = \pm 1$: by [Proposition 9.26](#), the eigenvalues of the periodic matrix $L(1) = L$ and of the antiperiodic one $L(-1)$. Both are real symmetric, so every band edge is real; spectral duality, below, shows that there are $2N$ of them, counted with multiplicity.

[Figure 9.6](#) shows $t(\lambda)$ for a six-particle ring: $N = 6$ bands, where it crosses the strip $|t| \leq 2$ monotonically, and $N - 1 = 5$ gaps, where it turns round outside it ([Corollary 9.30](#) proves this is general). The momenta keep changing; the band edges do not.

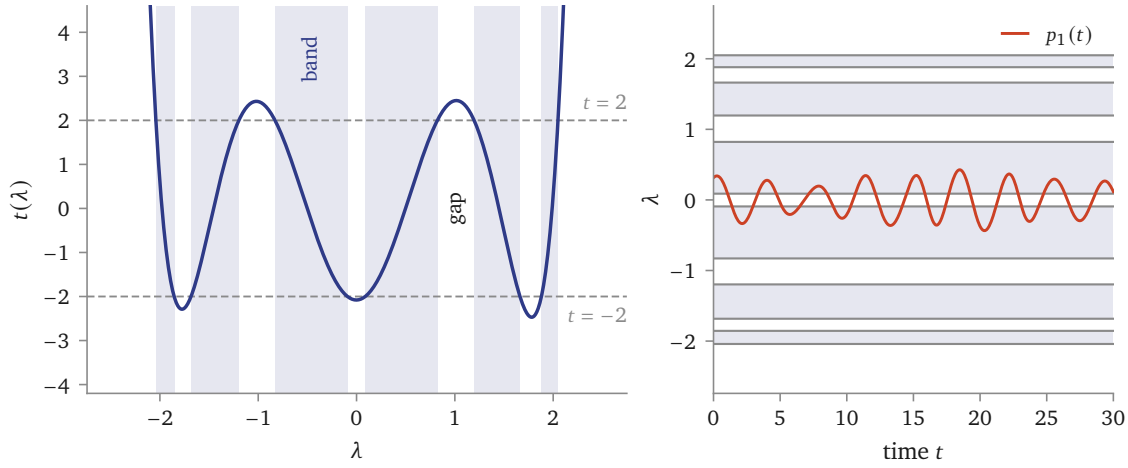


Figure 9.6: Left: $t(\lambda)$ for a state of the $N = 6$ ring, bands shaded. Right: the $2N$ band edges (the spectra of $L(\pm 1)$) along the motion, constant while $p_1(t)$ oscillates (`scripts/figures/toda_bands.py` asserts the edges constant to 10^{-8} and equal to the roots of $t \mp 2$).

At the equilibrium, all q_n equal and all $p_n = b$, every $a_n = 1$ and one finds $t(\lambda) = 2T_N((\lambda - b)/2)$, with T_N the Chebyshev polynomial: it touches ± 2 at every turning point, all gaps are closed, and the spectrum is the single band $[b - 2, b + 2]$ of the free tight-binding model. Outside the loci where a gap collapses, which have codimension two (Henrici and Kappeler, §2 · *cited*), all $N - 1$ gaps are open; a special excitation can leave some closed (a state repeating with a shorter period does).

Example 9.28 (All gaps closed: the equilibrium). At $q_1 = \dots = q_N$ and $p_1 = \dots = p_N = b$ every $a_n = 1$, and $L(z)$ is the free tight-binding model. Its Bloch waves are plane waves $\psi_n = e^{ikn}$, with $e^{ikN} = z$ and energy $\lambda = b + 2 \cos k$. Putting $\lambda - b = 2 \cos k$,

$$t(\lambda) = z + \frac{1}{z} = 2 \cos Nk = 2T_N\left(\frac{\lambda - b}{2}\right),$$

with T_N the Chebyshev polynomial, $T_N(\cos k) = \cos Nk$. It touches ± 2 at every turning point, so all $N - 1$ gaps are closed and the spectrum is one band $[b - 2, b + 2]$. This state is the

ring at rest in its centre-of-mass frame, drifting with speed b . Outside codimension-two loci every gap is open (Henrici–Kappeler, cited), but a special excitation need not open them all (for $N = 4$, a state repeating with period 2 has $t = d^2 - 2$, d the two-site trace, and two of its three gaps stay closed). A gap is the spectral trace of motion relative to the drift, as the actions below make quantitative. At this state the dc_k are dependent. By the cyclic symmetry every X_{c_k} moves all particles alike, and it changes no momentum, since $\sum_n \partial c_k / \partial q_n = 0$ (the c_k see only differences of positions) and the N terms are equal. So every flow is the drift, the X_{c_k} span one direction, and for $N \geq 2$ the dc_k cannot be independent.

The curve as a two-sheeted cover. Above each λ lie two points of Γ , z_+ and $z_- = 1/z_+$, which coincide exactly at the band edges. So Γ is two copies of the λ -plane (the *sheets*) glued along cuts joining the band edges in pairs; crossing a cut exchanges z and $1/z$ (Figure 9.7). With all gaps open, $2N$ branch points and N cuts give a closed surface with $N - 1$ handles: genus $N - 1$ (Babelon–Bernard–Talon, §6.2 · cited), the dimension of the tori of Theorem 9.11. The actions below say why.

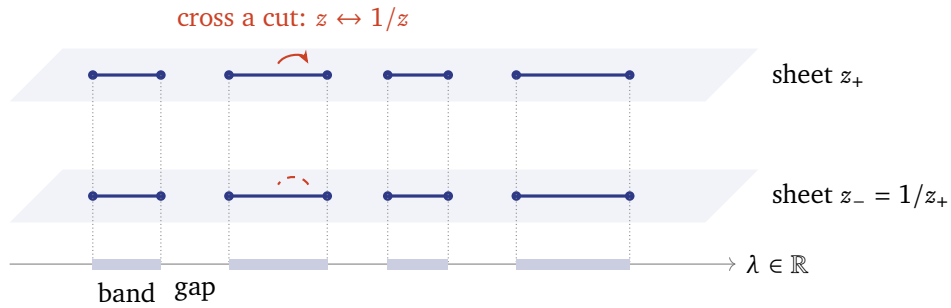


Figure 9.7: The spectral curve as a two-sheeted cover of the λ -plane, drawn over the real axis for $N = 4$. The cuts (thick) run along the bands and end at the $2N$ branch points, where $z_+ = z_- = \pm 1$; crossing a cut changes sheet. Two sheets glued along N cuts make a surface with $N - 1$ handles.

A warning on names. Babelon–Bernard–Talon exchange the letters: their λ is our z , their μ our λ , their $2t(\mu)$ our $t(\lambda)$, and their $n + 1$ particles our N (so their genus n is our $N - 1$).

Reading

Spectral duality: one curve, read two ways

Proposition 9.26 matched the two eigenvalue problems as sets. The polynomial identity is sharper.

Theorem 9.29 (Spectral duality). For $N \geq 3$ and all $\lambda \in \mathbb{C}$, $z \neq 0$,

$$\det(\lambda - L(z)) = t(\lambda) - \left(z + \frac{1}{z}\right) \quad (9.27)$$

The left side is the characteristic polynomial of the $N \times N$ Lax matrix, whose spectral parameter is z ; the right side is the characteristic equation of the 2×2 monodromy matrix, whose spectral parameter is λ . They define the same curve Γ .

Proof. Expand the determinant over permutations σ . The matrix $\lambda - L(z)$ is tridiagonal apart from its corners, $-a_N z$ at $(N, 1)$ and $-a_N/z$ at $(1, N)$. A term using the $(N, 1)$ corner and not the $(1, N)$ one must map $N \mapsto 1$ and, to stay within the band, $n \mapsto n + 1$ for every n : the N -cycle, of sign $(-1)^{N-1}$, whose term is $(-1)^{N-1} \prod_n (-a_n) z = -z$ by (8.11). Likewise the other corner alone gives $-1/z$, and every other term uses both corners or neither and does not involve z . So $\det(\lambda - L(z)) = A(\lambda) - (z + 1/z)$ with A independent of z . Fix λ and choose z with $z + 1/z = A(\lambda)$. Then $\det(\lambda - L(z)) = 0$, so λ is an eigenvalue of $L(z)$, so by Proposition 9.26 $z + 1/z = t(\lambda)$. Hence $A(\lambda) = t(\lambda)$. ■

The merged $N = 2$ matrix satisfies (9.27) too (scripts/check-l10-algebra.py, $N = 2, \dots, 5$, symbolic).

Two spectral parameters, one curve.

- z , the *Bloch–Floquet multiplier*, lives in the corners of the $N \times N$ matrix $L(z)$. Fix z : N points of Γ , the eigenvalues of $L(z)$. Γ is an N -sheeted cover of the z -plane.
- λ , the *energy* of the Bloch wave, is the spectral parameter of the 2×2 site matrices. Fix λ : 2 points of Γ , the eigenvalues $z, 1/z$ of $T(\lambda)$. Γ is a *two-sheeted* cover of the λ -plane.
- The eigenvalue of one Lax matrix is the spectral parameter of the other. The Flaschka normalisation was chosen so that this holds with no factor.

Figure 9.8 draws the N -sheeted reading over the unit circle $|z| = 1$: the familiar band diagram of a crystal. A vertical line meets it in N points, a horizontal line inside a band in 2.

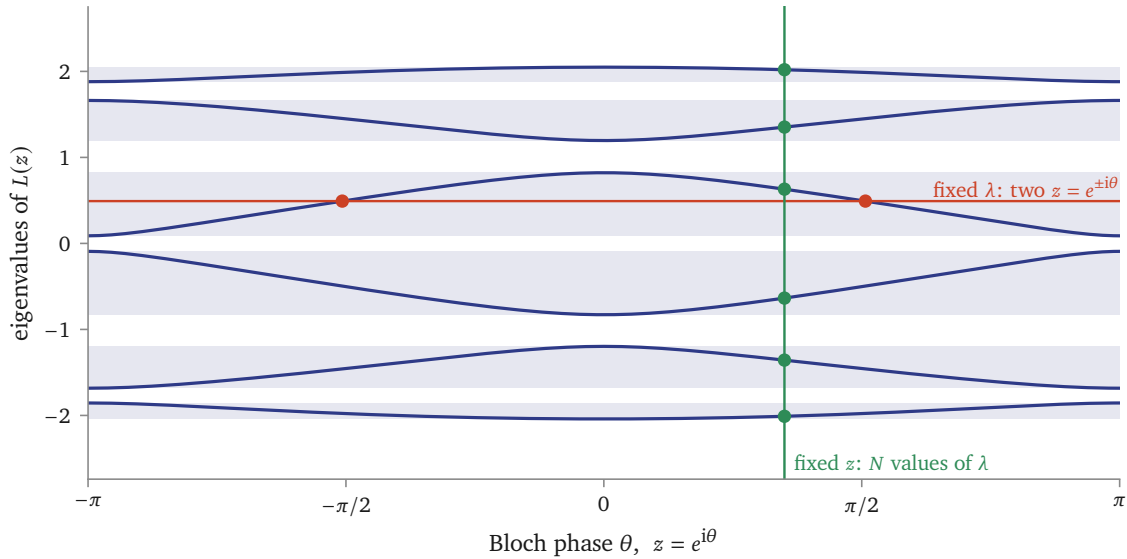


Figure 9.8: The state of Figure 9.6: eigenvalues of $L(e^{i\theta})$ against the Bloch phase. Each of the $N = 6$ branches sweeps one band. Fixed z (green): N values of λ ; fixed λ (red): two values of z (toda_bands.py asserts both, and (9.27) at sampled points).

Duality pays off at once.

Corollary 9.30 (Bands and gaps). (i) $\det(\lambda - L) = t(\lambda) - 2$, so the coefficients c_k and the integrals $F_k = \text{tr } L^k$ of § 8.6 determine each other, and the F_k are in involution. (ii) $t'(\lambda) \neq 0$

wherever $|t(\lambda)| < 2$. Hence the real line consists of exactly N bands, on each of which t runs monotonically from ± 2 to ∓ 2 , separated by $N - 1$ gaps, some of which may be closed.

Proof. (i) is (9.27) at $z = 1$, and Newton's identities express the power sums $\text{tr } L^k$ of the eigenvalues through the coefficients of the characteristic polynomial, and back. (ii) At a point of a band interior, $t(\lambda_0) = 2 \cos \theta$ with $z = e^{i\theta} \neq 1/z$, so the eigenvalue λ_0 of the Hermitian $L(z)$ is simple by Proposition 9.26, and it is a simple root of $t(\lambda) - 2 \cos \theta$ by (9.27). The count of bands follows, as below. ■

Proof of (ii) in full. Let $t(\lambda_0) = 2 \cos \theta$ with $0 < \theta < \pi$, and $z = e^{i\theta}$. Since $z \neq 1/z$, the eigenspace of $T(\lambda_0)$ for z is one-dimensional, so by Proposition 9.26 so is that of $L(z)$ for λ_0 . $L(z)$ is Hermitian, so λ_0 is a simple root of $\det(\lambda - L(z)) = t(\lambda) - 2 \cos \theta$, and $t'(\lambda_0) \neq 0$. So on each maximal open interval where $|t| < 2$ the polynomial is strictly monotone; the interval is bounded, since $t \rightarrow \pm\infty$, and $t = +2$ at one end and -2 at the other. Each such interval therefore contains exactly one solution of $t(\lambda) = 2 \cos \theta$ for each $\theta \in (0, \pi)$. There are exactly N solutions, the eigenvalues of $L(e^{i\theta})$, all simple, so there are exactly N such intervals: the bands. Between consecutive bands lie the $N - 1$ gaps, closed where two bands touch. ■

Reading

What z does and does not buy

An honest remark: z gives no new integrals. Expanding (9.27) with Newton's identities,

$$\text{tr } L(z)^k = \text{tr } L^k \quad (k < N), \quad \text{tr } L(z)^N = \text{tr } L^N + N \left(z + \frac{1}{z} - 2 \right).$$

The z -dependence is carried by $a_1 \cdots a_N$, a constant by (8.11): $L(z)$ holds exactly the integrals L had. Contrast Manakov's parameter in § 8.11, which recovered a missed integral.

Where the spectral parameter is indispensable. In the 2×2 formulation: all N integrals are the coefficients of one 2×2 trace, and without λ the site matrices carry nothing. In the bracket: (9.9) compares λ with μ , and without it there is no composition along the ring; Lecture 14 quantises exactly this, with $T(\lambda)$ operator-valued, RTT in place of (9.9), and $t(\lambda)$ generating a spin-chain Hamiltonian. And in the geometry: the actions of the tori of Theorem 9.11 are periods of the curve.

Action variables from the curve. When all $N - 1$ gaps are open, the curve has $N - 1$ closed loops that stay over the real axis (a closed gap collapses its loop to a point): above the j -th gap, both roots $z_{\pm}$ are real, and the loop runs along the gap on one sheet and back on the other, crossing between them at the two band edges (Figure 9.9). Call it α_j .

Proposition 9.31 (Actions from the gaps · cited). On a regular torus of Theorem 9.11 with all $N - 1$ gaps open, the loops α_j correspond to a basis γ_j of cycles of the torus T^{N-1} in the sense of Lecture 5, and the action variables (6.1) of the relative motion are

$$I_j = \frac{1}{2\pi} \oint_{\alpha_j} \lambda \, d \ln z = \frac{1}{\pi} \int_{\text{gap } j} \text{arccosh} \frac{|t(\lambda)|}{2} \, d\lambda, \quad j = 1, \dots, N - 1. \quad (9.28)$$

(Flaschka and McLaughlin 1976; in this normalisation, Henrici and Kappeler 2008, Definition 3.1 and Appendix A. The separated variables behind it are Babelon–Bernard–Talon, §§6.5–6.6.)

The second form follows by integrating by parts, with $\ln |z_{\pm}| = \pm \operatorname{arccosh}(|t|/2)$ over a gap and α_j oriented so that $I_j \geq 0$. A closed gap has zero action; a wide gap means a large amplitude. The proof needs separated variables and the Riemann surface of Γ , which this course does not build. `scripts/check-l10-algebra.py` checks (9.28) for $N = 4, 5$ near equilibrium, where $H - N - F_1^2/2N = \sum_j \nu_j I_j$ to leading order, $\nu_j = 2 \sin(\pi j/N)$ (relative error below 10^{-5}), and the example below does $N = 2$ by hand.

Example 9.32 (The two-particle ring). Take $N = 2$, with $x = q_1 - q_2$ and $p_x = \frac{1}{2}(p_1 - p_2)$, a canonical pair, and the mean momentum $\bar{p} = \frac{1}{2}(p_1 + p_2)$ (not P , the permutation operator). Then $H = \bar{p}^2 + E$ with relative energy $E = p_x^2 + 2 \cosh x \geq 2$: a particle in a cosh well. From (9.4), with $p_1 p_2 = \bar{p}^2 - p_x^2$,

$$t(\lambda) = (\lambda - p_1)(\lambda - p_2) - e^x - e^{-x} = u^2 - E, \quad u = \lambda - \bar{p}.$$

There is one finite gap, $|u| < \sqrt{E - 2}$, where $t < -2$, and there $\operatorname{arccosh}(|t|/2)$ is the $x_0 \geq 0$ with $2 \cosh x_0 = E - u^2$. So the right side of (9.28) is $\frac{1}{\pi} \int x_0(u) du$: the area of the region $\{u^2 + 2 \cosh x \leq E\}$ in the (u, x) -plane, divided by 2π . Rename u as p_x : it is the area enclosed by the orbit in the (x, p_x) -plane over 2π , which is $\frac{1}{2\pi} \oint p_x dx$. On the slice $Q = \text{const}$ we have $p \cdot dq = 2\bar{p} dQ + p_x dx = p_x dx$, so this is the action of Definition 6.1. The same region, sliced two ways.

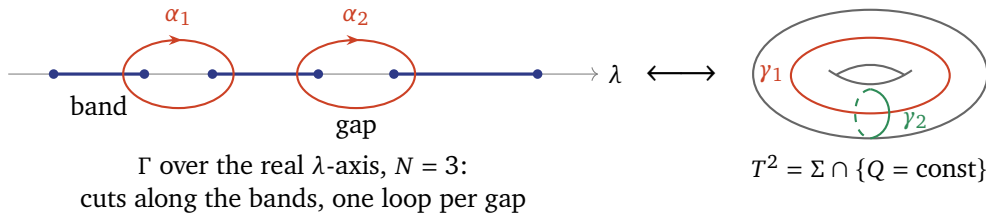


Figure 9.9: Left: for $N = 3$, two gaps and two real loops α_1, α_2 on the curve, each along a gap on one sheet and back on the other. Right: the relative torus T^2 of Theorem 9.11 with a basis of cycles. Proposition 9.31 (cited) matches the two.

This is the sense in which the genus of Γ equals the dimension of the tori: each open gap is a handle of the curve and a circle of the torus, and its width measures how hard that circle is excited. Theta functions on Γ then give the motion itself in closed form (Babelon–Bernard–Talon, §6.4 · *cited*), which we do not pursue.

Summary.

- *All integrals at once.* The monodromy $T = L_N \cdots L_1$ and the transfer matrix $t = \operatorname{tr} T$ (Definition 9.1); its coefficients $c_1, \dots, c_N$ include $H = \frac{1}{2}c_1^2 - c_2$ (9.5), and they are Lecture 8's integrals reorganised, $\det(\lambda - L) = t(\lambda) - 2$ (9.6).
- *Tensor legs.* $\{A_1, B_2\}$ has $(ac), (bd)$ entry $\{A_{ab}, B_{cd}\}$; different legs commute as matrices (Remark 9.3).
- *One identity for one site.* The quadratic Sklyanin bracket (9.9) composes along an ultralocal chain (Theorem 9.7), and its trace gives $\{t(\lambda), t(\mu)\} = 0$ (Corollary 9.8).
- *Toda.* The site matrix satisfies it with $r = -P/\lambda$ (Proposition 9.9); with the Vandermonde argument, the ring is Liouville integrable (Theorem 9.10), and its regular level sets are $T^{N-1} \times \mathbb{R}$ (Theorem 9.11). Goal G1 is reached.

- Next. Part II begins: [Lecture 10](#) constructs the Heisenberg chain's eigenstates by Bethe's guess, [Lecture 14](#) quantises this lecture's three moves into commuting transfer matrices, and [Lecture 15](#) diagonalises them and recovers the Bethe equations.

Tutorial. Exercise Sheet 9.

References

Babelon, Bernard and Talon, *Introduction to Classical Integrable Systems*, ch. 2–3 (the Sklyanin bracket) and ch. 6 (the periodic chain, its 2×2 monodromy matrix and its spectral curve); Faddeev and Takhtajan, *Hamiltonian Methods in the Theory of Solitons*, Part I ch. 1 (the classical r -matrix and the transition to the quantum case); H. Flaschka and D. W. McLaughlin, *Prog. Theor. Phys.* 55 (1976) 438 (the actions from the gaps). **Conventions.** Babelon–Bernard–Talon exchange the letters λ and z , and conventions for the sign of r differ between all three; see the third caveat.